File permissions in Linux

Project description

I work with the research team in a large organization as a security professional. My job is to ensure users on this team are authorized with the appropriate permissions. This helps keep the system secure.

I was tasked to examine existing permissions on the file system. I'll determine if the permissions match the authorization that should be given. If they do not match, I'll modify the permissions to authorize the appropriate users and remove any unauthorized access. To complete this task, I performed the following tasks:

Check file and directory details

The following code demonstrates how I used Linux commands to determine the existing permissions set for `projects/` directory in the file system.

```
researcher2@978cdfc576e8:~$ ls -al  projects/
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:55 ..
-rw--w---- 1 researcher2 research_team   46 Mar 16 00:08 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 16 00:08 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar 16 00:08 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 16 00:08 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_t.txt
researcher2@978cdfc576e8:~$
```

The first line shows the command I entered `ls -la projects/.` The ls is used for listing the content of the directory. `-la` is a flag, `-a` to display the hidden files like `.project_x.txt` and `-l` to display the details like date and permissions, the two of them combined will be `-la`. The full command shows that there are five files and one directory.
The permissions on `drafts` are listed in line six. They are drwx - -x- - -:

- d = directory

The next three characters are permissions for the user(rwx)

- r = read permission.
- w= write permission.

- x= execute permission.

Then there are the group permission(--x)

- -= there is no read permission.
- -= no write permission.
- x= there is an execute permission.

Last the other users permissions(- - -)

- -= read not allowed.
- -= write not allowed.
- -= execute not allowed.

Hence the permissions on this directory are:

- User = read, write, execute
- Group = execute
- Other = none

Describe the permissions string

The 10-character string can be deconstructed to determine who is authorized to access the file and their specific permissions. The characters and what they represent are as follows:

- **1st character**: This character is either a `d` or hyphen (`-`) and indicates the file type. If it's a `d`, it's a directory. If it's a hyphen (`-`), it's a regular file.
- **2nd-4th characters**: These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for the user. When one of these characters is a hyphen (`-`) instead, it indicates that this permission is not granted to the user.
- **5th-7th characters:** These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for the group. When one of these characters is a hyphen (`-`) instead, it indicates that this permission is not granted for the group.
- **8th-10th characters:** These characters indicate the read (`r`), write (`w`), and execute (`x`) permissions for other. This owner type consists of all other users on the system apart from the user and the group. When one of these characters is a hyphen (`-`) instead, that indicates that this permission is not granted for other.

For example the `.project_x.txt` file has (-rw--w---) permissions. The first character is the hyphen indicating that this is a regular file. The 2nd-4th characters are (rw-), indicating read and write permissions for the user, but not execute. Next, 5th-7th are -w-, meaning the group only has write permission. Finally, the 8th-10th characters are ---, meaning the other users have no permissions on this file.

Change file permissions

The organization determined that other shouldn't have write privileges to any file. Therefore, the `project_k.txt` should restrict the access.

```
researcher2@978cdfc576e8:~/projects$ chmod o-w project_k.txt
researcher2@978cdfc576e8:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:55 ..
-rw--w---- 1 researcher2 research_team   46 Mar 16 00:08 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 16 00:08 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 16 00:08 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_t.txt
researcher2@978cdfc576e8:~/projects$
```

I used the chmod command to change the permission. `o` stands for other, `-w` for removing the write (`w`) permission. Then list the content with `ls -al` to see the effect.

Change file permissions on a hidden file

The policies restrict the access for hidden files to read only for owner and group.

```
researcher2@978cdfc576e8:~/projects$ chmod 440 .project_x.txt
researcher2@978cdfc576e8:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:55 ..
-r--r----- 1 researcher2 research_team   46 Mar 16 00:08 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar 16 00:08 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 16 00:08 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_t.txt
researcher2@978cdfc576e8:~/projects$
```

Here I used the octal numbers to represent read-only for user by 4, read-only to group by 4, and no permissions to other by 0.

Change directory permissions

The organization states that only the `researcher2` user should be allowed to access the drafts directory and its contents. (This means that only `researcher2` should have execute privileges).

```
researcher2@978cdfc576e8:~/projects$ chmod 700 drafts/
researcher2@978cdfc576e8:~/projects$ ls -al
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:08 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar 16 00:55 ..
-r--r----- 1 researcher2 research_team   46 Mar 16 00:08 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar 16 00:08 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar 16 00:08 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar 16 00:08 project_t.txt
researcher2@978cdfc576e8:~/projects$
```

The first line shows the command I used which is `chmod 700 drafts/`. The chmod to change the permission on the directory and the `700` is octal number stands for 7=4(r)+2(w)+1(x) and the `00` says that there is no permissions for group and other.

Summary

In this activity I changed the permissions in order to comply with the organization's policies for securing the system. First, I used `ls -al` to list the content for reviewing the permissions. Then the `project_k.txt` file had write permission for other, so I removed it. Next, I granted read-only permission for `.project_x.txt` for both the user and the group. Finally, for the drafts directory I only kept all permission (read, write, execute) to the user and removed the execute permission from the group.

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